
EMERGENCY ROUTES

Real-Time Route Optimization for Emergency Medical Services

Polytechnic University of Valencia (UPV)

Bachelor's Degree in Data Science

Project III

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1. Abstract

Conventional navigation systems do not account for real-time road closures, extreme weather conditions, or large-scale urban events. In emergency situations, this forces ambulances to recalculate routes reactively, losing critical minutes. *Emergency Routes* is an integrated decision-support system, developed in collaboration with the Generalitat Valenciana, that addresses this limitation through two complementary modules. The first is a route optimization engine that incorporates road restrictions as exclusion polygons in OpenRouteService, with seven predefined scenarios (Fallas, Marathon, DANA, PATRICOVA, among others) and support for multi-casualty incidents under the START/SHORT doctrine and the MANV dispatch model. The second is an automated triage system that transcribes emergency calls using Whisper and classifies their severity with a fine-tuned mBERT model (**bert-base-multilingual-cased**), reaching a macro F1 of 85.4% and perfect recall in the *Severe* category. The system has been evaluated on real data from the province of Valencia, including the DANA event of October 29, 2024 as a maximum stress test, demonstrating that integrating artificial intelligence into the CICU emergency workflow can structurally reduce the operational latency of the service.

2. Introduction and Motivation

On October 29, 2024, the Valencian Community was hit by one of the most devastating floods in recent European history. The catastrophe caused by the DANA storm led to massive road closures, communication blackouts, and the displacement of thousands of people, leaving emergency services unable to navigate a city whose infrastructure had become unrecognizable within hours. Ambulances were diverted onto blocked streets they were unaware of until they reached them. Calls to the coordination center accumulated faster than operators could answer them. Response times, in moments where every minute is decisive, stretched dangerously. And none of this was a human failure — it was a tool failure.

2.1 The Problem: Current Systems

The Emergency Medical Service in Valencia is well structured under normal conditions, but it shows its limits when facing scenarios of high urban complexity. Standard commercial navigation tools compute optimal routes based on distance and average travel time, but they do not account for real-time road closures, weather conditions, or unpredictable traffic patterns generated by events such as the Valencia Marathon, the Fallas, or public demonstrations, among others.

Beyond routing, there is a critical bottleneck for emergency surges: the call itself. CICU operators must simultaneously handle incoming calls, classify the severity of the incident, and coordinate the deployment of the appropriate units, all in real time and under extreme pressure. The lack of automated triage support means that the urgency of a call depends, in part, on human availability and subjective judgment at the worst possible moment.

2.2 Our Proposal

This project proposes an integrated, AI-based decision-support system for optimizing emergency medical services in the province of Valencia, developed in collaboration with the Generalitat Valenciana. The project is organized around two complementary modules.

The first is a route optimization system: a dynamic routing interface capable of recalculating, in real time, the trajectory of SAMU and SVB units when a road segment is blocked, incorporating restrictions specific to each scenario, such as Fallas closures or marathon corridors.

The second module is audio-based triage, an automated system that processes calls received at 112, transcribes audio to text, and predicts the severity level of the patient, supporting the CICU operator in resource prioritization and faster, more objective decision-making.

2.3 Scope and General Objective

The system relies on real operational data from the SAMU and SVB fleets in Valencia, the provincial hospital network, and scenario-specific datasets for those events most likely to disrupt normal service operations.

The goal is not to replace the judgment of emergency professionals, but to give them better information, faster, so that when the next crisis comes, the tools are equal to the situation.

The fundamental question driving this project is simple: *how many minutes could be saved if the CICU always had the fastest route, the most critical patient identified, and the most precise estimated time of arrival?*

3. Related Work and State of the Art

In the Valencian Community, emergency medical management is organized around 112, the single European emergency telephone number and the main instrument the Generalitat provides for citizens to request urgent assistance. When a call is received, the operator gathers essential incident information, identifies which services must intervene, and activates the appropriate resources: medical, fire, police, or civil protection, depending on the type and location of the emergency.

Medical coordination falls under the CICU (*Centro de Información y Coordinación de Urgencias*, Information and Emergency Coordination Center), which supervises emergency situations, activates the necessary protocols, and centralizes information until the incident is resolved. The CICU mobilizes two main types of healthcare units: the SAMU (*Servicio de Atención Médica de Urgencia*), which operates as an advanced life support unit equipped with medical and nursing staff and specialized equipment; and the SVB units (*Soporte Vital Básico*, Basic Life Support), staffed by Emergency Medical Technicians and intended for emergencies that do not require immediate medical intervention, or for stabilizing the patient until advanced support arrives. The decision of which resource to mobilize, and when, depends on a correct severity assessment from the first moment.

However, the DANA of October 29, 2024 exposed the limits of this model in the face of a disaster of such scale. At 17:00, a peak of 2,438 simultaneous calls was recorded, and many of them could not be answered due to 112 saturation. In total, on that day the system received 19,821 calls and managed 4,770 incidents — figures that overwhelmed normal operational capacity. But, beyond the saturation of the call system, the collapse of the road network added an additional layer of inoperability: available ambulances could not reach those affected, because conventional dispatch systems did not account for the real state of the roads in real time.

Conventional navigation systems such as Google Maps are designed for normal traffic conditions. In a scenario like the one described above, these systems cannot incorporate real-time closures or recalculate routes avoiding flooded areas. In that context, an ambulance guided by Google Maps could be sent down an impassable road, losing critical minutes turning around and recalculating the route.

4. Methodology and Technical Development

The system developed integrates, as mentioned above, two complementary modules over a common data foundation: route optimization and audio triage. Both share the same data infrastructure and have been designed to operate in a coordinated way within the real emergency management workflow of the CICU.

4.1 Data Used

Seven heterogeneous data sources were gathered for the project, all referring to the province of Valencia. Of these, four directly feed the system modules in their current state, while three were collected, cleaned, and analyzed within the framework of a predictive modeling objective (ETA with weather and traffic variables) that was ultimately deferred as a continuation proposal, as detailed in the limitations section. Table 1 summarizes all sources, their origin, and their integration status.

4.1.1 Hospitals and Ambulances

The hospital dataset, once cleaned, contains 18 hospital centers distributed across the province. The initial inspection revealed 27 records corresponding to ambulance bases erroneously included in the source file, which were removed after cross-checking the coordinates against the unit inventory. The final dataset contains no missing values, duplicate coordinates, or observations outside the provincial bounding box. Exploratory analysis reveals a markedly unbalanced geographical distribution: most hospitals are concentrated in the Valencia metropolitan area, while inland departments such as Requena-Utiel and Xàtiva-Ontinyent each have a single center. The mean distance to the nearest neighbor is 6.2 km, with maximum values exceeding 50 km for the most isolated inland hospitals. These facilities are precisely the most vulnerable to road closures, since they have a very limited number of alternative routes.

The ambulance inventory comprises 71 operational units distributed across the province. Each unit is characterized by its type, distinguishing between SAMU (Advanced Life Support, 22

Table 1: Data sources used in the project.

Source	Origin	Use in the system
Hospital directory	Generalitat Valenciana (provided by Yulia Karpova)	Destination nodes of the routing model
Ambulance inventory (SAMU and SVB)	Generalitat Valenciana (provided by Yulia Karpova)	Origin nodes of the routing model
Road restriction layers (7 scenarios)	Valencia City Council / GVA / OSM	Closure polygons and risk zones for the routing engine
Emergency call dataset	Synthetic	Training of the triage model
Traffic intensity 2023–2025	Generalitat Valenciana	Development for future projects
River flows and reservoirs	Júcar Hydrographic Confederation	Development for future projects
Weather data (2025–2026 and DANA day)	AEMET	Development for future projects

units) and SVB (Basic Life Support, 49 units), as well as by its operational shift: 24 hours, 12 hours daytime, or 12 hours nighttime. Coordinate-based inspection identified multiple units sharing the same physical base, a critical factor for the model, since a single road closure could simultaneously disable several co-located units.

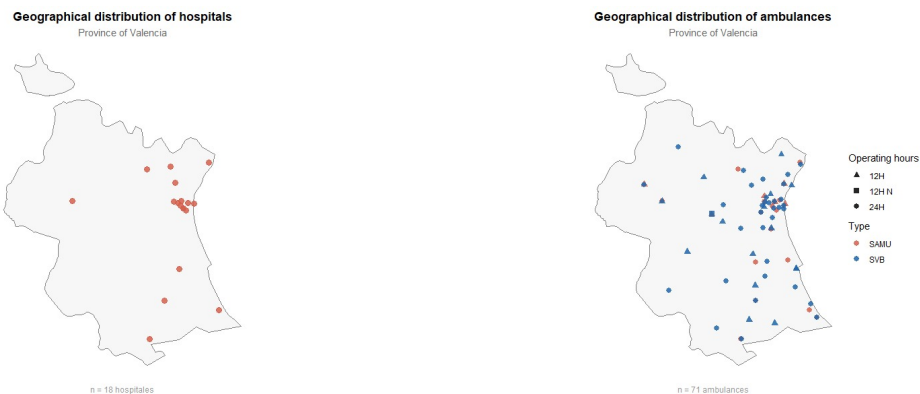


Figure 1: Geographical distribution of hospitals (left, $n = 18$) and ambulances (right, $n = 71$) in the province of Valencia.

4.1.2 Road Network, PATRICOVA, and Restriction Scenarios

The provincial road network has been enriched with the flood hazard classification of the PATRICOVA plan, which assigns each segment a risk level between 1 (maximum) and 6, plus level 7 corresponding to geomorphological hazard with no quantified return period. The analysis shows that segments with the highest flood risk (level 2 or lower) are disproportionately rural and secondary roads — precisely those providing access to inland hospitals.

The route optimization module operates over a collection of seven geospatial layers in CSV format, each representing a distinct road restriction scenario that can be activated in the interface at dispatch time. Each file encodes the polygons of the affected road segments, so they can

be passed directly to the `avoid_polygons` parameter of OpenRouteService for the engine to treat them as impassable zones during route computation. The construction process for these layers varies by scenario, although the final output is homogeneous: a CSV in which each row corresponds to a street polygon with its coordinates in EPSG:25830 (UTM zone 30N), which the system automatically reprojects to WGS84 before sending to ORS.

The closure layer for the Fallas of Valencia was obtained from a shapefile published by the Valencia City Council with the official itineraries and closures of the event. Since the file already contained the polygons of the closed streets in vector format, processing was limited to coordinate system reprojection and CSV export.

For the Valencia Marathon, Semana Marinera, San Vicente Procession, and Ofrenda de Fallas, the input data were only the official routes published on their respective web pages, available as lines rather than road polygons. The construction process was the same in all four cases: the official route was traced in QGIS as a line layer, a spatial intersection was performed with the complete Valencia street network (OSM) to identify the crossed segments, a fixed-width buffer was applied to convert the linear geometries into surface polygons, and finally the layer was exported to CSV.

The layer for the DANA of October 29, 2024 was built from the official water footprints published by the Generalitat Valenciana, which delimit the areas effectively flooded during the event. These footprints were intersected in QGIS with the provincial street network to extract only the road segments inside the flooded zones; the same buffering process was applied and the result exported to CSV. This event, which recorded precipitation of up to 710.8 mm in a single day at the Turís station, with 10 of the 37 active AEMET stations under red alert, constitutes the maximum-severity scenario of the system.

The PATRICOVA extreme risk layer was obtained by filtering exclusively those segments with hazard level 1 and 2, corresponding to the highest risk categories. Unlike the other scenarios, this layer does not represent a specific closure but a structural map of road risk that can be activated preventively in the face of intense rainfall episodes.

4.1.3 Emergency Call Dataset

For training the severity classification model, a synthetic dataset of 504 emergency calls was generated (168 per class), evenly distributed by language (Spanish and Valencian) and severity level (mild, severe, and moderate). The dataset is perfectly balanced by design: being fully synthetic, the class distribution was controlled to prevent the model from learning to always predict the majority class. Synthetic generation was necessary because real 112 calls are confidential information, and operational records of the emergency service were not accessible. The dataset was partitioned into train (70%) / validation (15%) / test (15%) with stratification, ensuring that the three severity levels are represented in the same proportions across each subset.

4.1.4 Data Collected for Future Development

In addition to the sources integrated into the system, three additional datasets were collected, cleaned, and analyzed; they were originally intended for the ETA predictive model with weather and traffic variables. The AEMET meteorological data, collected for both the 2025–2026 period and the DANA event day, contained more than 79% missing values for pressure and sunshine-hour variables, due to limited instrumental availability at the stations. The Generalitat Valenciana traffic intensity data for 2023–2025 covered multiple road segments and required extracting coordinates embedded in text fields and normalizing date formats. The hydrographic data from the Júcar Hydrographic Confederation (CHJ) provided information on basin reservoir levels. As discussed in the limitations section, the impossibility of building a generalizable model from a single extreme event led the team to defer this objective as a continuation proposal. Nevertheless, the work of collecting and preparing these data is documented and available in the project repository, so future developments can resume it without starting from scratch.

4.2 Route Optimization with Closures

The route optimization module is the central decision-making layer of our system. Its goal is to determine, in real time and under road closure scenarios, which ambulance should be dispatched to an incident and which route to follow to minimize the *Total Operational Latency* (defined as the sum of the ambulance-to-incident time and the incident-to-hospital transfer time).

The system models the road network of the province of Valencia as a directed, weighted graph, where intersections are nodes and road segments are edges with a cost based on travel time. To solve the routing problem, OpenRouteService (ORS) is used as the computation engine, which internally applies algorithms such as Dijkstra or A* over the OpenStreetMap network. This allows obtaining routes optimized over real road infrastructure without the computational cost of maintaining a custom graph.

The main contribution of the module is the ability to inject road closures as geometric restrictions directly into the engine *before* any route is computed. Closed or flooded segments are defined as polygons, drawn manually on the map or imported in files with UTM coordinates (EPSG:25830), which the system automatically reprojects to WGS84. These polygons are passed to ORS through the `avoid_polygons` parameter, which forces the algorithm to treat them as impassable zones from the start, instead of detecting the obstacle reactively once the ambulance is already underway. This proactive approach is what differentiates the system from conventional navigation tools.

When an incident is registered, the system does not evaluate all available ambulances. First it filters units by operational availability (only free ambulances on active shift at that moment are considered). It then computes the Haversine distance between each candidate and the incident, selecting the eight closest. For each of these, it queries ORS and obtains the actual travel time under the active closures. The ambulance with the lowest adjusted time is selected for dispatch. The same logic is applied in parallel to select the destination hospital, filtered by incident severity: critical cases are sent to hospitals with full surgical capacity, while less severe ones can be diverted

to health centers.

The system also incorporates a Multi-Casualty Incident mode based on the START/SHORT triage protocol. Victims are classified by severity (red, yellow, green, black) and the system automatically computes the number of SAMU and SVB units required according to MANV protocols, prioritizing SAMU for critical patients and SVB for evacuation transport. In scenarios with multiple ambulances dispatched, the system also activates a hospital dispersion logic to avoid saturating a single emergency department: when the number of dispatched units reaches or exceeds a configurable threshold (set by default to 5), patients are distributed among the two or three nearest hospitals instead of being concentrated in a single one. This threshold has been set conservatively under the assumption that, below that volume, a single hospital can absorb demand without risk of collapse; it can be adjusted depending on the operational context or the actual capacity of receiving centers. To preserve the system’s global response capacity against simultaneous emergencies, an additional saturation cap of 70% of the available fleet is applied at any moment: units exceeding that cap remain in reserve and are logged as a pending second dispatch wave. Finally, to minimize latency in repeated calculations, the system implements a route cache that stores previous ORS responses, automatically invalidated when new closure polygons are introduced.

4.3 Audio Detection, Transcription, and Triage System

The audio detection module is the entry point of the system, responsible for capturing, processing, and classifying emergency calls in real time. Its objective is to minimize the interval between call reception and severity-level assignment, removing the dependency on a human operator for initial classification and allowing resource dispatch to begin in parallel with telephone attention.

The system receives the call audio as an input file in standard formats (**mp4**, **m4a**, or **wav**) and processes it directly with Whisper. This design fits naturally with the CICU telephone systems, which already record calls in these formats. The file is passed to OpenAI’s Whisper model in its **small** variant. Whisper is a Transformer encoder–decoder speech recognition model, pre-trained by OpenAI on 680,000 hours of audio across 99 languages. Internally it works in two parts: first, the encoder analyzes the audio by decomposing it into its frequencies and how they evolve over time — similarly to how the human ear distinguishes sounds — and extracts a mathematical representation of the sound content. Then the decoder takes that representation and generates the corresponding text word by word. The **small** variant (244 million parameters) was chosen as the best balance between quality and speed for Spanish and Valencian: the **tiny** variant is faster but makes more errors with accents and medical terminology, while **medium** and **large** are significantly slower without a sufficient improvement for short emergency calls. Transcription operates with reduced precision disabled to guarantee compatibility in environments without GPU, although in the presence of GPU the process is practically instantaneous.

The transcribed text is then delivered to the severity classifier. For this task we use models from the BERT family, which are language models based on the Transformer architecture. Unlike previous models that read text left-to-right or right-to-left, BERT reads the entire sentence at once and learns the meaning of each word as a function of all the others around it. This is achieved

through a mechanism called *self-attention*, which for each word computes how important every other word in the text is to understand its meaning. For example, in the sentence “*the patient is not breathing and is unconscious*”, the model learns that “unconscious” is strongly related to “patient” and to “not breathing”, and uses that relationship to build a more precise representation of the whole. These models are especially suitable for classifying emergency calls because they capture the complete context of what is said, not just isolated words, which allows distinguishing between a mild and a severe situation even when the described symptoms are similar.

The specific classifier used is **bert-base-multilingual-cased** (mBERT), fine-tuned over a corpus of 504 synthetic emergency calls, balanced across three classes: mild, severe, and moderate. mBERT was chosen over other alternatives for three concrete reasons. First, compared to monolingual models such as BETO (Spanish only), mBERT natively handles Valencian without prior translation, having been pre-trained on Wikipedia in 104 languages including Catalan and Romance languages close to Valencian. Second, compared to XLM-RoBERTa, mBERT achieved equivalent validation results (macro F1 0.853, accuracy 0.855) with lower computational cost, which makes it more suitable for an operational emergency environment where latency matters. Third, mBERT uses WordPiece tokenization, which splits words into smaller, more frequent fragments, allowing it to handle uncommon medical terms well by decomposing them into recognizable parts. The model tokenizes the transcription, applies inference in evaluation mode with gradients disabled, and produces a probability distribution over the three levels via softmax. The class with the highest probability determines the assigned triage level, which is communicated to the route optimization module together with the full transcription to condition both ambulance selection and destination hospital.

The system also incorporates an automatic medical-context search via the public Spanish Wikipedia API. Starting from the clinical terms extracted from the transcription, filtered with a stopword list in Spanish and Valencian, the summary of the most relevant article is retrieved and included in the report generated for the ambulance team before its arrival at the incident scene. This mechanism requires no API key and operates asynchronously to avoid introducing latency in the main classification.

5. Model Evaluation and Iteration

To select the final model, a systematic hyperparameter search was performed, exploring 6 configurations: 2 architectures (**bert-base-multilingual-cased** and **xlm-roberta-base**) $\times$ 3 combinations of learning rate and epochs (lr=2e-5/ep=2, lr=3e-5/ep=2, lr=5e-5/ep=3). The guiding metric was macro F1 on validation, which weights all three classes equally and especially penalizes errors on the *severe* class. The winning model was **bert-base-multilingual-cased** (mBERT) with lr=5e-5 and 3 epochs, achieving a macro F1 of 0.853 and an accuracy of 0.855 on validation, tied with **xlm-roberta-base** under the same hyperparameters. mBERT was selected as the final model because it is computationally lighter while maintaining the same performance, which is a priority in an emergency system where classification latency is critical. The remaining configurations performed notably worse, particularly those with lower learning rate (lr=2e-5 or 3e-5) or fewer epochs, confirming that the chosen configuration is the most adequate for this

problem.

The search was performed manually and systematically over the tested data, supported by the comparison chart shown in Figure 2.

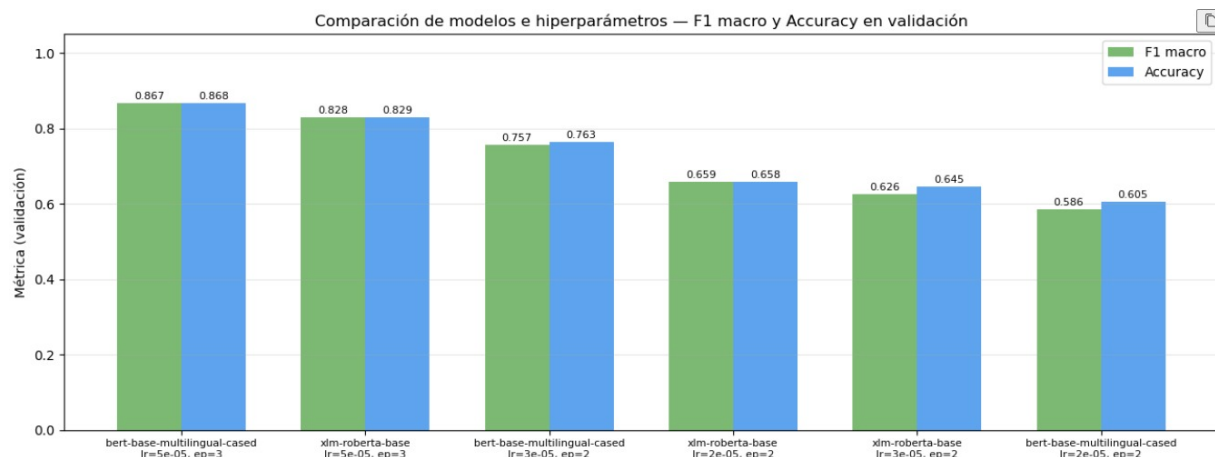


Figure 2: Accuracy and macro F1 comparison across tested models.

BERT Model Results

After evaluating the best model **bert-base-multilingual-cased** (mBERT) on a test set of 76 instances, distributed across the three classes (25 Mild, 25 Moderate, and 26 Severe), the results show solid and balanced performance: 85.5% accuracy, 87.4% macro precision, 85.3% macro recall, and 85.4% macro F1.

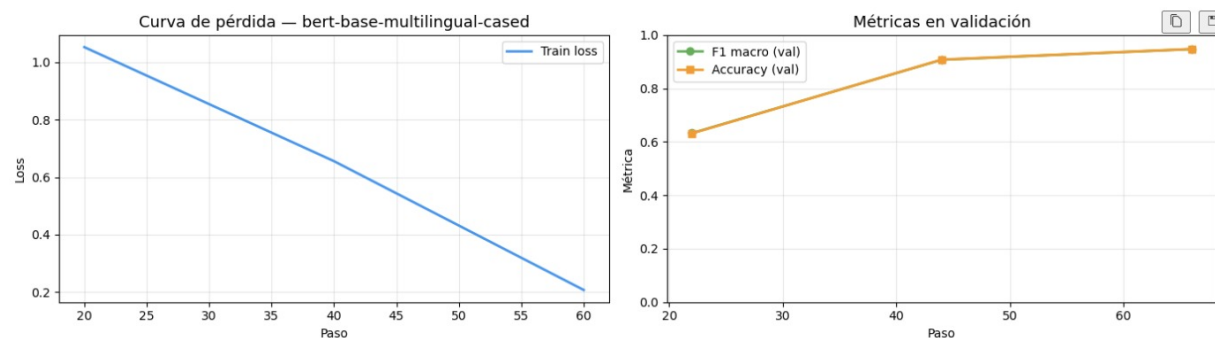


Figure 3: Evolution of train loss, macro F1, and accuracy per training step.

The training curves confirm that the fine-tuning process worked correctly. Training loss decreases continuously from 1.1 to approximately 0.35 along the 80 recorded steps, meaning the model progressively learned to distinguish among the three severity levels and did not get stuck in any local minimum. The fact that the curve descends smoothly without abrupt jumps also indicates that the learning rate was appropriate and that training was stable. In parallel, both macro F1 and validation accuracy grow consistently along training, starting from values around 0.3 and 0.4 respectively in the first steps and stabilizing above 0.85 from step 65 onward. The fact that validation metrics continue to rise while training loss decreases, without a gap opening between them, indicates that the model is not overfitting — that is, what it learns generalizes to calls not seen during training.

Per-class analysis reveals performance differences with clinically relevant interpretation. The *moderate* category is the one the model classifies most reliably, reaching a perfect recall of 1.00 and a precision of 0.90: all 26 moderate cases in the test set were identified correctly, with none going undetected. This means the model never underestimates a moderate emergency.

The *mild* category shows the highest precision of the three, at 1.00: every time the model predicts a call as mild, it is correct. However, its recall of 0.72 indicates that some mild cases are classified with a higher severity — specifically, 1 as moderate and 6 as severe. From a clinical point of view, this kind of error is conservative and preferable to the opposite, since it is better to send extra resources than to leave without priority attention a case that turns out to be urgent.

The *severe* category obtains a recall of 0.92, correctly classifying 23 of its 25 cases. The 2 remaining errors are confused with *moderate*, never with *mild*. This behavior is clinically acceptable: a severe case classified as moderate will still receive immediate attention, never being left unattended.

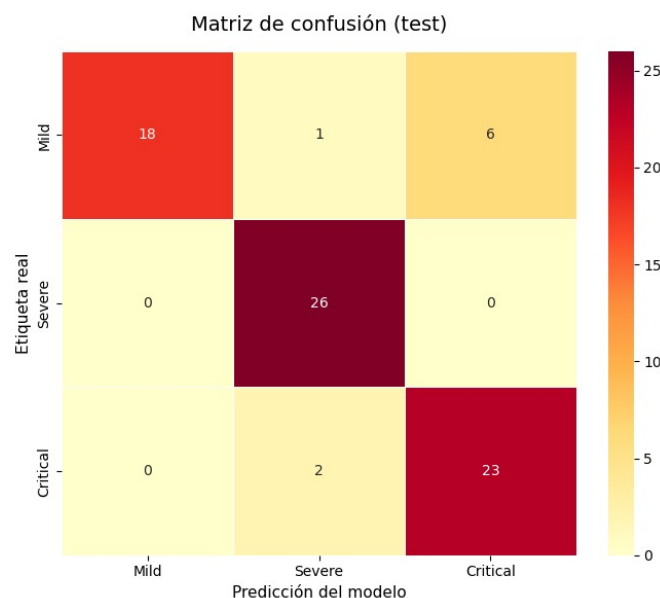


Figure 4: Confusion matrix of the best model (test set).

The confusion matrix clearly illustrates this pattern. All 26 moderate cases are classified correctly, with a perfect recall of 1.00. The main errors concentrate in the mild category, where 6 instances are predicted as severe and 1 as moderate. In the severe category errors are minimal, with only 2 instances confused with moderate. Most importantly, from a clinical point of view, there is no case in which a severe emergency is classified as mild — which would be the most dangerous error of the system.

The test on the example audio reinforces these results qualitatively. Given the transcription “*My mother fainted in the street, it was at Carral 38, I need an ambulance as soon as possible*”, the model assigns the *severe* level with 64.0% confidence, versus 30.0% for moderate and 5.9% for mild. That the confidence is not higher makes sense: a syncope may have benign causes or be the onset of a severe cardiovascular event, and the model reflects that ambiguity by assigning 30% to moderate. The final decision to escalate to the higher level is the desirable behavior in a

real emergency system, where when in doubt it is always preferable to send more resources than to fall short.

5.1 System Architecture and Deployment

One of the key objectives of this project was to make the system accessible from any device, anywhere, without requiring any local installation. To achieve this, we designed a fully cloud-based architecture using three free services: Netlify, Firebase, and Hugging Face Spaces.

Frontend: Netlify. Both web portals (the citizen portal and the dispatch portal) are static HTML files hosted on Netlify. Netlify provides a public URL for each deployment, meaning any user can open either portal directly in a browser (on a phone, tablet, or computer) without installing anything. Every time a new version of the files is uploaded via Netlify Drop, the deployment updates automatically in seconds.

Real-time Communication: Firebase Firestore. The main challenge of having two separate portals running on different devices was making them communicate in real time. We solved this using Firebase Firestore, a NoSQL cloud database by Google. When the citizen submits an alert, the system writes a document to the `avisos` collection in Firestore. This document contains the transcription, the severity level classified by the AI, the GPS coordinates, and the current status. The dispatch portal listens to that collection using Firestore’s `onSnapshot` listener, which pushes any new document to the operator instantly (no polling, no page refresh needed). Once the operator dispatches an ambulance, the system updates that same document with the assigned unit and the estimated arrival time. The citizen portal, which is also listening to its own document, receives that update immediately and notifies the user.

AI Model: Hugging Face Spaces. The AI pipeline (Whisper for transcription and the fine-tuned mBERT model for severity classification) is hosted on Hugging Face Spaces using a Docker container. This means the model runs permanently in the cloud and is accessible via a public API endpoint. The citizen portal sends the recorded audio to that endpoint, receives the transcription and classification, and forwards the result to Firebase. The main limitation of this approach is that Hugging Face Spaces puts the container to sleep after a period of inactivity on the free tier. The first request after a sleep period takes slightly longer while the container wakes up. A production deployment would require a dedicated server to guarantee response times.

6. Results and Value Assessment

The fundamental value of *Emergency Routes* lies in its ability to transform static data and theoretical models into operational, real-time tools. Unlike conventional navigation platforms, our system enables proactive decision-making, anticipating obstacles before the emergency unit starts moving. Below we present a qualitative evaluation of the system through real-scenario simulations, an analysis of its impact, and the technical limitations identified.

System Demonstration: Evaluation Through Real Scenarios

To validate the robustness of the route optimization module, the system has been subjected to several stress tests using the spatial restriction CSV files developed in the methodology.

1. Scheduled Urban Events and Closure Density (Fallas Scenario)

As shown in the captures of the system’s operational interface, the Fallas scenario poses a complex geometric challenge due to the proliferation of hundreds of small polygons (marquees, monuments, and closed streets) distributed across the city.

In the demonstration without active closures, we simulate a *Severe* emergency on *Carrer del Comte d’Altea* (L’Eixample). The system automatically assigns an advanced life support unit (SAMU A041 General) with an arrival time of 6.9 minutes. The direct evacuation route to the Hospital Clínico de Valencia takes just 3.9 minutes, resulting in a total operational latency of 11 minutes.

When the Fallas CSV is injected (with 781 active closures in red), the situation changes radically. Although the ambulance’s arrival time at the incident remains stable (6.8 minutes), the direct evacuation route to the north is completely blocked by Falla groupings. The system processes these obstacles through OpenRouteService’s `avoid_polygons` parameter and instantly draws an alternative route (dashed blue line) that goes around the closed blocks. This detour raises the hospital transfer time to 6.5 minutes, placing the total time at 13 minutes. The value of this demonstration is enormous: in a real environment, without this system, an ambulance carrying a severe patient on board would have attempted the direct route (3.9 min), would have found the street blocked by a marquee, and would have lost vital minutes turning back and blindly searching for alternative routes.

2. Flexibility and Scalability: Additional Scenarios and Custom CSVs

The main strength of this optimization module lies in its technological agnosticism towards the nature of the obstacle. In addition to the scenarios cited, the system is prepared to receive scenarios such as the Valencia Marathon route or several processions acting as dynamic barriers. The routing engine does not care *why* a street is closed — it only needs its spatial delimitation. This gives the platform absolute versatility: the operator can upload any CSV file with closure polygons instantly (accidents, demonstrations, roadworks) and the system will adapt the routes of the entire fleet in real time, without requiring developer intervention.

3. Multi-Casualty Incident (MCI) and MANV Dispatch Doctrine

The most distinctive technical and operational advance of the project is the algorithmic translation of the medical triage doctrine (START/SHORT) and the German dispatch model (MANV) into executable code. Unlike linear dispatch systems, the MCI mode processes the emergency under the following clinical parameters:

- **START triage and alert levels.** The operator enters victims via four counters (Red/Immediate, Yellow/Urgent, Green/Mild, Black/Deceased). The system automatically contextualizes the

volume: Level 1 (up to 10 victims), Level 2 (11–25, recommending establishment of an Advanced Medical Post in the log), and Level 3 (>25 victims, suggesting activation of the Health Emergency Plan).

- **Role assignment by unit type (MANV).** The system differentiates vehicle roles in the algorithm. It first dispatches the closest SAMU units to set up the Advanced Medical Post (PSA) (with a doctrinal cap of 4 SAMU for stabilization). It then deploys the SVB units needed to build the “evacuation wheel”.
- **Dynamic fleet calculation.** Strict resource arithmetic is applied. If 3 Red, 4 Yellow, 5 Green, and 1 Black victims are reported, the system computes:
 - 3 SAMU (one per Red). If Reds were to exceed the 4-SAMU cap, the surplus would be absorbed by SVBs.
 - 2 SVB for the Yellows (applying the 1:2 optimization ratio of the German doctrine).
 - Greens are referred to the Advanced Medical Post without immediate transport, and Blacks do not consume transport resources, automatically generating a legal notice for the on-duty court.
- **70% provincial fleet cap.** To guarantee system resilience, the algorithm never dispatches more than 70% of the total available ambulances. If the disaster requires more, it dispatches the “first wave” up to the cap and alerts the remaining pending units, preventing the rest of Valencia from being left uncovered.
- **Intelligent hospital dispersion.** If the global computation requires fewer than 5 units, the system sends all of them to the fastest hospital. However, the moment more than 5 ambulances are dispatched, the algorithm activates a dispersion logic: it evenly distributes Red and Yellow patients across the 2 or 3 hospitals with the shortest transfer time, ensuring that the emergency department of a single center is not overwhelmed.

To evaluate the algorithmic precision of all these rules operating simultaneously, we executed a simulation in the city center (*Carrer de Colón*) with a total of 16 victims: 5 Immediate (Red), 6 Urgent (Yellow), 4 Mild (Green), and 1 Deceased (Black).

The system demonstrated a precise execution of the integrated doctrine:

1. It automatically catalogued the event as Level 2 – MCI, suggesting installation of the Advanced Medical Post.
2. It applied the doctrinal cap by dispatching the maximum of 4 SAMU units for the Advanced Medical Post (PSA).
3. It assigned 4 SVB units for the evacuation wheel (calculated by adding 1 SVB for the excess Red critical patient and 3 SVB to cover the 6 Yellow patients under the 1:2 ratio).
4. Requiring a total deployment of 8 ambulances, the algorithm exceeded the saturation threshold (5 units) and activated hospital dispersion, drawing evacuation routes to different hospitals in the city to avoid overwhelming a single emergency department, generating a simultaneous logistical deployment visible in the cartographic interface.

Beneficiaries

The proposed system targets three levels of beneficiaries with distinct but interconnected needs. The direct user is the CICU, whose coordination staff would have an integrated tool to optimize unit dispatch in real time, receive an automated first estimate of patient severity, and compute more precise arrival times under adverse conditions. Second, the healthcare teams of the SAMU and SVB units themselves would benefit from receiving routes already recalculated before starting the deployment, instead of facing unexpected road closures under the pressure of a waiting patient. The final beneficiary is the patient, for whom every minute of reduction in response time can mean the difference between a timely intervention and an irreversible outcome.

Expected Impact

In the absence of real CICU dispatch operational data, the time reduction the system would deliver in production cannot be quantified exactly. However, the operational impact can be argued qualitatively as being structural along at least two dimensions. The first is the elimination of reactive recalculation time: in the current workflow, when an ambulance encounters a road closure en route, the driver must stop, look for an alternative, or contact the CICU, losing critical minutes. With the proposed system, the alternative route would be computed before the unit leaves the base, eliminating that latency entirely. The second dimension is improved dispatch prioritization in multi-casualty incidents. The combination of automated audio triage with ETA prediction would allow the CICU to assign the most appropriate units (SAMU or SVB) to the most critical patients simultaneously, rather than processing calls sequentially under time pressure.

Limitations

It is important to transparently note the current limitations of the project, both technical and methodological, to provide a realistic picture of its maturity.

Although the system is conceptually designed to manage extreme climatic disasters — such as the recent DANA scenario (based on the PATRICOVA plan) — empirical execution has revealed technological restrictions of both hardware and third-party services. Theoretically, the system's goal in vast flooded metropolitan areas (such as Paiporta or Catarroja) is to automatically redirect units via secondary roads to accessible peripheral hospitals (such as La Ribera or Xàtiva), avoiding dispatching ambulances along impassable highways and thus safeguarding the fleet.

However, regarding the route optimization module, the public OpenRouteService API imposes a strict load limit on the number of vertices and restriction polygons (`avoid_polygons`) that can be included in a single query. In a massive catastrophe scenario such as the DANA, where thousands of segments were simultaneously flooded, the public version of the API is insufficient to process the full exclusion cartography at once, causing the request to fail. This shows that, to achieve real operational deployment at regional scale, a paid API subscription, or ideally a self-hosted OpenRouteService instance on Generalitat servers, would be required to overcome this load limit.

Finally, one of the project’s initial objectives was to develop a predictive model of flood probability based on data from AEMET, the Generalitat Valenciana, and the Júcar Hydrographic Confederation. However, during the model design phase a fundamental problem was identified: the DANA of October 29, 2024 constitutes the *only* recorded event reaching flood levels of that magnitude in the province of Valencia. This means the available dataset contains a single positive case of the target variable, making it impossible to build a robust train/test partition or guarantee the model’s generalization capacity. Given this limitation, the team decided to reorient the objective: instead of a predictive model, the real DANA data were used as a simulation scenario within the route optimization module, and the construction of a flood prediction model has been deferred as a continuation proposal for future projects, conditional on the availability of additional event records.

7. Conclusions and Legacy

Lessons Learned

The development of *Emergency Routes* has confirmed something that is not always obvious at the start of a project of this kind: the emergency routing problem is not just an algorithmic problem. The implementation of the routing engine, the dispatch logic, and the MANV protocols required real technical effort and design decisions that were not trivial. But the most costly and decisive part of the work came before all of that: building the road restriction layers, understanding PATRICOVA formats, converting UTM coordinates to WGS84, managing inconsistent OpenStreetMap geometries. In a project of this kind, data quality conditions the final result far more than algorithmic sophistication.

A second, equally relevant lesson is that real medical protocols translate directly and precisely to software logic, provided they are understood properly before writing a single line of code. The implementation of the MANV protocol was not possible until we genuinely understood how START/SHORT triage works in practice. That understanding, which came in part thanks to contact with sector professionals, changed specific design decisions that would otherwise have produced a technically correct but operationally useless system.

The audio triage module brought its own lessons. Building the training dataset revealed how blurry the semantic boundary is between a *moderate* call and a *severe* one when only the voice transcription is available: the same symptom description can shift category based on tone, the speaker’s age, or contextual details that text does not capture. Despite this, the BERT model achieved perfect recall on the severe class, showing that even with limited data it is possible to build a system that prioritizes patient safety above all. The most valuable lesson here was accepting that a triage model does not need to be perfect on every class — it needs to never fail on the one that matters most.

Dissemination and Continuation Plan

The full source code of the project, together with technical documentation and the datasets not subject to confidentiality restrictions, will be released in an open GitHub repository to

facilitate the replicability of the work. The repository will include the two system modules (route optimization, audio triage), cleaning and exploratory analysis scripts in R, and the instructions needed to reproduce the results in a local environment.

In addition to the public repository, the complete project will be shared with researchers Yulia Karpova and M.^a Fulgencia Villa, collaborators on the project during its development, who will evaluate the possibility of offering the system to the Generalitat Valenciana or to the Health Emergency Service of the Valencian Community (SESCV) as an operational support tool. This transfer route constitutes the most realistic path for the project to have an impact beyond the academic sphere.

From a technical standpoint, the most natural continuation proposal would be integration with real-time data sources. The DGT traffic API or OpenStreetMap alerts would allow closure polygons to update automatically rather than being loaded manually by the coordinator. That integration would transform the system from a decision-support tool into an autonomous response system for road disruptions — a qualitative leap that would require validation with real SAMU dispatch historical data. A second continuation line, conditional on the availability of future data, is the flood probability prediction model. As discussed in the limitations section, the DANA of October 29, 2024 is the only event of that magnitude recorded in the province of Valencia, making the construction of a generalizable model unfeasible. If new episodes are recorded in the coming years, or if historical records from other comparable hydrographic basins become accessible, that model could become the fourth module of the system, closing the full cycle from risk prediction to operational response.

Alignment with the Sustainable Development Goals

The project aligns directly with three of the Sustainable Development Goals identified in the initial phase of the work. **SDG 3 (Good Health and Well-Being)** is the most immediate: reducing the operational latency of the emergency system has a direct effect on survival in critical situations. **SDG 11 (Sustainable Cities and Communities)** connects with urban resilience to disasters, the scenario that motivated the project from its origin. And **SDG 13 (Climate Action)** closes the argument: the DANA was not an exceptional event but a signal of what is to come, and an emergency system that is not prepared for extreme climatic conditions is not prepared for the future.

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9. Appendix (code, external tools, and additional tables)

Table of data sources used

Source	Origin	Use in the system
Hospital directory	Generalitat Valenciana (provided by Yulia Karpova)	Destination nodes of the routing model
Ambulance inventory (SAMU and SVB)	Generalitat Valenciana (provided by Yulia Karpova)	Origin nodes of the routing model
Road restriction layers (7 scenarios)	Valencia City Council / GVA / OSM	Closure polygons and risk zones for the routing engine
Emergency call data	Synthetic	Triage model creation
Traffic intensity 2023–2025	Generalitat Valenciana	Development for future projects
River flows and reservoirs of the different basins	Júcar Hydrographic Confederation	Development for future projects
Meteorological data (2025–2026 and DANA day)	AEMET	Development for future projects

Table 2: Data sources used in the project (appendix).

Hospital data sample

tipo	nombre	direccion	codigo_postal	municipio	longitud	latitud	comarca	provincia	cod_comarc	DPTOCRC	departamento
hospital	HOSPITAL UNIVERSITARIO DE SAGUNTO	AV. RAMON Y CAJAL, S/N	46520	Sagunt/Sagunto	-0.2319574	39.67486	el Camp de Morvedre	València/Valencia	12	4	04. SAGUNTO
hospital	H. CLINICO DE VALENCIA	AV BLASCO IBAÑEZ, 17	46010	València	-0.3618910	39.47863	València	València/Valencia	15	5	05. VALENCIA - CLINICO
hospital	H. LA MALVA-ROSA	ISABEL DE VILLENA, 2	46011	València	-0.3254072	39.47758	València	València/Valencia	15	5	05. VALENCIA - CLINICO
hospital	H. DE LLIRIA	PARATGE CABEÇO DE L'AGUILA, CV-35, S/N	46160	Lliria	-0.6191397	39.64194	el Camp de Túria	València/Valencia	11	6	06. VALENCIA ARNAU LLIRIA

Figure 5: Sample of the hospital dataset.

Ambulance data sample

Tipo	Horario	Provincia	Nombre	Departamento	Dirección	Longitud	Latitud
SAMU	24H	Valencia	A053 Manises	23	Aparcamiento, Av. de la Generalitat Valenciana, 50, 46940 Manises, Valencia	-0.5	39.5
SAMU	24H	Valencia	A032 Alfahuir	5	Carrer d'Ismael Merlo, 46020 València, Valencia, España	-0.4	39.5
SAMU	24H	Valencia	A033 Campanar	7	Av. Campanar, 23 Valencia	-0.4	39.5
SAMU	24H	Valencia	A041 General	9	Av. Tres Forques, 98, 46018 Valencia	-0.4	39.5
SAMU	24H	Valencia	A055 Silla	10	Calle Vicenta Gomar, Comare, 9 bajo	-0.4	39.4

Figure 6: Sample of the ambulance dataset.